Assignment 1

Course Code:	IT2B (Network)
Course Name:	Network Administration
Assignment:	CIDR Assignment

Materials and Resources

Textbook:	Lammle, Todd. (2021). CompTIA Network+ Study Guide (5e). Sybex.

Assignment Description

In this assignment, you will solve CIDR-related problems.

Assignment Steps

Activity Steps

1. Binary-Decimal Number Conversion **(12 marks)**

   a. Convert the following decimal numbers to binary.

   88 = 0 1 0 1 1 0 0 0
   241 = 1 1 1 1 0 0 0 1
   99 =0 1 1 0 0 0 1 1
   42 =0 0 1 0 1 0 1 0
   169 =1 0 1 0 1 0 0 1
   311 = cannot be solved with 8 bits

   b. Convert the following binary numbers to decimal.

   01011100 = 92
   00111100 = 60
   11110000 =240
   10101011 =171
   11001011 =203
   01111101 =125

2. Subnetting Class A, B, and C Networks **(30 marks)**

   Subnet the following networks (using 2^n = number of subnets) – figure out the subnet mask required, the first 3 usable ranges, and the number of hosts per subnet.

   3.0.0.0/8 network (with 6 subnets) **(5 marks)**
   Borrow n=3bits 8+3= /11, subnet mask: 255.224.0.0
   Total host bit=32-11 =21bits
   Total IPs per subnet 2^21=2,097,152
   Usable Host per Subnet 2^21-2=2,097,150

	N I.D	First IP	Last IP	Broadcast IP	
1	3.0.0.0	3.0.0.1	3.31.255.254	3.31.255.255	
2	3.32.0.0	3.32.0.1	3.63.255.254	3.63.255.255	
3	3.64.0.0	3.64.0.1	3.95.255.254	3.95.255.255	

192.168.7.0/24 network (with 11 subnets) **(5 marks)**
Borrow n=4bits 24+4= /28, subnet mask: 255.255.255.240
Total host bit=32-28 =4bits
Total IPs per subnet 2^4=16
Usable Host per Subnet 2^4-2=14

N I.D	First IP	Last IP	Broadcast IP
192.168.7.0	192.168.7.1	192.168.7.14	192.168.7.15
192.168.7.16	192.168.7.17	192.168.7.30	192.168.7.31
192.168.7.32	192.168.7.33	192.168.7.46	192.168.7.47

131.19.0.0/16 network (with 2 subnets) **(5 marks)**

Borrow n=1bits 16+1= /17, subnet mask: 255.255.128.0

Total host bit=32-17 =15bits
Total IPs per subnet 2^15=32,768
Usable Host per Subnet 2^15-2=32766

N ID	First IP	Last IP	Broadcast IP
131.19.0.0	**131.19.0.1**	**131.19.127.254**	**131.19.127.255**
131.19.128.0	**131.19.128.1**	**131.19.255.254**	**131.19.255.255**

111.0.0.0/8 network (with 15 subnets) **(5 marks)**
Borrow n=4bits 8+4= /12, subnet mask: 255.240.0.0
Total host bit=32-12 =20bits
Total IPs per subnet 2^20=1048576
Usable Host per Subnet 2^20-2=1048574

Network ID	First IP	Last IP	Broadcast IP
111.0.0.0	**111.0.0.1**	**111.15.255.254**	**111.15.255.255**
111.16.0.0	**111.16.0.1**	**111.31.255.254**	**111.31.255.255**
111.32.0.0	**111.32.0.1**	**111.47.255.254**	**111.47.255.255**

177.2.0.0/16 network (with 21 subnets) **(5 marks)**
Borrow n=5bits 16+5= /21, subnet mask: 255.255.248.0
Total host bit=32-21 =11bits
Total IPs per subnet 2^11=2048
Usable Host per Subnet 2^11-2=2046

Network ID	First IP	Last IP	Broadcast IP
177.2.0.0	177.2.0.1	177.2.7.254	177.2.7.255
177.2.8.0	177.2.8.1	177.2.15.254	177.2.15.255
177.2.16.0	177.2.16.1	177.2.23.254	177.2.23.255

220.13.42.0/24 network (with 3 subnets) **(5 marks)**

Borrow n=2bits 24+2= /26,subnetmask:255.255.255.192
Total host bit=32-26 =6bits
Total IPs per subnet 2^5=64
Usable Host per Subnet 2^6-2=62

Network ID	First IP	Last IP	Broadcast IP
220.13.42.0	220.13.42.1	220.13.42.62	220.13.42.63
220.13.42.64	220.13.42.65	220.13.42.126	220.13.42.127
220.13.42.128	220.13.42.129	220.13.42.190	220.13.42.191

3. Difficult Subnet Examples **(8 marks)**

Subnet the following network (using 2^n = number of subnets) – figure out the subnet mask required, the first and last 3 usable ranges, and the number of hosts per subnet.

101.0.0.0/8 network (with 1300 subnets)

Borrow n=11bits 8+11= /19, subnet mask: 255.255.224.0
Total host bit=32-19 =13bits
Total IPs per subnet 2^13=8192
Usable Host per Subnet 2^13-2=8190

First 3 usable ranges:
101.0.0.1 - 101.0.31.254
101.0.32.1 - 101.0.63.254
101.0.64.1 - 101.0.95.254

Last 3 usable ranges:
101.255.160.1 - 101.255.191.254
101.255.192.1 - 101.255.223.254
101.255.224.1 - 101.255.255.254

4. Identifying the Broadcast ID of a Subnet. **(7 marks)**

    a. Identify the broadcast ID for the 2nd usable subnet in all exercises in Question 2.

3.0.0.0/8: 2nd Subnet range: 3.32.0.0 to 3.63.255.255
Broadcast ID: 3.63.255.255
192.168.7.0/24: 2nd Subnet range: 192.168.7.16 to 192.168.7.31
Broadcast ID: 192.168.7.31
131.19.0.0/16: 2nd Subnet range: 131.19.128.0 to 131.19.255.255
Broadcast ID: 131.19.255.255
111.0.0.0/8: 2nd Subnet range: 111.16.0.0 to 111.31.255.255
Broadcast ID:111.31.255.255
177.2.0.0/16: 2nd Subnet range: 177.2.8.0 to 177.2.15.255
Broadcast ID: 177.2.15.255
220.13.42.0/24: 2nd Subnet range: 220.13.42.64 to 220.13.42.127
Broadcast ID: 220.13.42.127

Identify the broadcast ID for the 2nd usable subnet in Question 3.
101.0.0.0/8 (1300)

2nd Subnet Range: 101.0.32.0 to 101.0.63.255
Broadcast ID: 220.13.42.127

5. Identify whether the following hosts are on the same subnet (or not). **(6 marks)**
   a. 192.168.7.62/28 and 192.168.7.70/28 =NO
   b. 88.91.0.113/11 and 88.95.45.1/11 = YES
   c. 4.0.0.6/16 and 4.0.5.2/16 = YES
   d. 155.19.161.2/22 and 155.19.141.7/22= NO
   e. 201.251.199.21/26 and 201.251.199.55/26 = YES
   f. 181.14.22.7/25 and 181.14.22.102/25 = YES

6. Supernetting Class C Networks **(2 marks)**

   a. Figure out the subnet mask required for the following network that will be supernetted.

   192.168.88.0/24
   192.168.89.0/24
   192.168.90.0/24
   192.168.91.0/24
   Answer=255.255.252.0

   b. Figure out the broadcast ID for the supernetted network in a.
   Answer= 192.168.91.255

Rubric

See the marks awarded for each section outlined in the steps/instructions.

This assignment is out of /65.